# AI Usage Acknowledgment - C-Shell Project

**Declaration of Originality:** I designed the core architecture, data structures, and logic flow of this shell. I utilized AI (Google Gemini) primarily as an interactive tutor and documentation tool to outline the project timeline, understand complex POSIX C concepts, inter-process communication, and file descriptors. Instead of generating black-box code, I explicitly prompted the AI to explain the high-level flow, break down the syntax, and verify my logic. I fully understand and can explain all code present in my submission.

## Key Consultations & Actual Prompts Used

**1. Project Planning and Structural Layout** Before writing any code, I used the AI to break down the dense project specification into a manageable development timeline and file structure, explicitly requesting that it not write the code for me.

- **Exact Prompt Used:** *"dont give me code and layout a timeline for me"*

**2. Grasping the Execution Flow & IPC Architecture** To ensure I understood how my shell handled multiple processes and pipes simultaneously, I asked the AI to trace the data flow using real-world examples before moving forward.

- **Exact Prompt Used:** *"ok now give me overall flow of the execute file. give using examples, for example what the user types, what happens to the input, and what is output and how is it calculated"*

**3. Line-by-Line Syntax Understanding (Redirection & Multiplexing)** When implementing the complex logic required to redirect standard output to multiple files simultaneously (the C2/C3 requirements), I required the AI to explain the underlying system calls (`dup2`, `fork`, `tmpfile`) rather than just accepting the logic.

- **Exact Prompt Used:** *"do explain all the changes made line by line syntax by syntax, and why have we decided to go with a particular logic by giving examples"*
- **Exact Prompt Used:** *"explain line by line what is happening, what each variabel and syntax means and what the code does"*

**4. Clarifying Custom Functions & Pointers** I used the AI to verify my understanding of specific C mechanics, such as how passing by pointer works when resolving system paths via `$PATH`.

- **Exact Prompt Used:** *"what does this do esolve_command_path(args[0], resolved_path)"*

**5. Understanding Specification Requirements (Frecency Engine)** Before implementing the Frecency algorithm for the hop command, I asked the AI to explain the conceptual goal of the requirement without immediately jumping into code.

- **Exact Prompt Used:** *"first tell me what the frequency thing in hop is. i just knew that hop was cd command customised. this frequency thing explain to me, what the functions do in overall flow."*

## Full Chat Transcript

The complete record of AI assistance, including all architectural brainstorming, debugging sessions, and line-by-line explanations referenced above, is documented and verifiable at the following link:

https://share.gemini.google/NiQhBdHu83fJ